

Hamiltonian Dynamic Parser as a Foundation for Hypergraph Knowledge Representation

A Technical Analysis

Generated by Claude Opus 4.6, prompted by Rob Freeman

March 8, 2026

1 1. Overview and Thesis

This analysis examines how the Hamiltonian dynamic parser embodies principles that are deeply relevant to—and in some ways more principled than—the attention allocation and knowledge representation mechanisms in hypergraph-based AI systems like OpenCog Hyperon and its AtomSpace knowledge graph. The central thesis is that the parser’s energy-minimizing substitutional process generates both “cognitive objects” (parse constituents with substitution classes) and “cognitive distance” (the energy landscape over decompositions) as emergent products of the process itself, rather than requiring them as pre-specified inputs. This is a property that Hyperon’s architecture aspires to but has not yet fully realized.

2 2. Energy Minimization and the Hamiltonian Analogy

2.1 2.1 The Parser’s Energy Formula

The energy function is:

$$E = -\log(n_{\text{subs}}) + \log(\text{consensus}) - \log(n_{\text{full_length}})$$

Three terms:

- $-\log(n_{\text{subs}})$: More substitutes lower the energy. A span replaceable by many alternatives occupies a clearly defined functional role.
- $+\log(\text{consensus})$: Higher consensus *raises* energy. A parse constituent that constrains its context too tightly is not a productive compositional unit—diverse contexts indicate a good compositional unit.
- $-\log(n_{\text{full_length}})$: More full-length substitutes lower the energy, rewarding structures that produce substitutes matching the span’s own word count.

2.2 2.2 Mapping to Hypergraph Energetics

In OpenCog’s ECAN (Economic Attention Networks), each Atom carries Short-Term Importance (STI) and Long-Term Importance (LTI), updated via nonlinear dynamical equations. STI spreads along HebbianLinks recording co-activation. The parser’s energy formula computes a

local Hamiltonian for each possible parse constituent. Where ECAN assigns importance to individual Atoms and spreads it along pairwise links, the parser computes importance (negative energy) for *structured compositions*—entire binary tree decompositions. This is fundamentally more expressive:

- **ECAN:** Importance flows between individual nodes along edges. The structure of associations is fixed by the HebbianLinks.
- **Parser:** “Importance” (low energy) is computed for *candidate structures*—proposed ways of decomposing a span into left and right constituents, where each constituent carries its own substitution class. The structure itself is what is being evaluated.

The parser performs attention allocation over *structural hypotheses*, not just individual items. Each candidate parse is a hypothesis about how meaning composes, and the energy function selects among them.

2.3 The Hamiltonian Interpretation

In classical mechanics, a Hamiltonian describes total energy as a function of configuration. The system evolves toward energy minima. The CKY bottom-up process implements exactly this: it enumerates configurations (all binary splits of all spans, shortest-first) and selects the minimum-energy configuration. The “configuration space” is the set of all possible binary tree decompositions, and the dynamics drive toward the decomposition maximizing substitutional support while maintaining contextual openness.

This is a richer formulation than what AtomSpace currently provides. In AtomSpace, knowledge is represented as a static hypergraph with truth values and attention values. The parser suggests that a more dynamic formulation—where the *structure of the graph itself* is the variable being optimized—could be more powerful.

3 Emergent Distance and Emergent Objects

3.1 The Core Insight

Neither “cognitive objects” nor “distances between them” are inputs to the system. Both emerge from the substitutional process.

Objects emerge as substitution classes. When the parser processes the span “my hat”, it finds substitutes for “my” (e.g., “your”, “his”, “the”) and substitutes for “hat” (e.g., “coat”, “bag”, “car”). These substitution classes *are* the system’s cognitive objects—they define what “my” and “hat” mean in this context. Crucially, the substitution class for “hat” in “my hat” may differ from the substitution class for “hat” in “wear a hat” because different contexts yield different substitutes.

Distance emerges as energy difference. Two parse constituents are “close” if they produce overlapping substitution classes—if the same words can fill the same structural role. They are “distant” if their substitution classes are disjoint. This distance is never explicitly computed as a metric; it is implicitly encoded in the energy landscape. A split producing constituents with rich, distinct substitution classes has low energy (good); one producing impoverished or overlapping classes has high energy (bad).

3.2 3.2 Contrast with Pre-Specified Distances

In traditional distributional semantics and embedding-based approaches, distance is computed between pre-existing vector representations. The parser uses similarity computations as *tools within* the process, but the meaningful distance—the one determining parse structure—emerges from the full energy minimization.

This is relevant to Hyperon because one of the unsolved problems in Hyperon’s attention allocation is exactly this: how should the system determine which Atoms are “close” and should be co-activated? ECAN uses HebbianLinks recording co-activation history—a *symmetric* measure. The parser demonstrates that association should be *asymmetric* (see Section 4), and that the relevant notion of “closeness” is not between individual items but between *structural roles*.

3.3 3.3 Emergent Structure in Hyperon’s MeTTa

MeTTa, Hyperon’s core language, is designed for graph transformations. The parser’s process of building up parse constituents from shorter spans is itself a kind of graph transformation: each cached subparse is a node in an implicit graph, and the CKY process creates new nodes (longer spans with substitution classes) from existing nodes (shorter spans). In MeTTa, transformation rules are explicitly written; in the parser, the “rules” are implicit in the energy function and the substitutional evidence.

A productive synthesis would be to implement the parser’s energy-minimizing substitutional process *as* MeTTa graph transformations over a Distributed AtomSpace. Each parse constituent becomes an Atom; each substitution relationship a typed Link; the energy function encoded as attention values. The CKY ordering becomes a priority queue over MeTTa transformations: shorter-span transformations having higher priority.

4 4. Asymmetric Containment Scoring as a Model for Association

4.1 4.1 The Containment Score

The scoring uses:

$$\text{score}(A \rightarrow B) = \frac{|\text{contexts}(A) \cap \text{contexts}(B)|}{|\text{contexts}(A)|}$$

This is asymmetric—it measures what fraction of the *candidate’s* contexts are also contexts of the *target*. A small, specific word (few contexts) can score highly against a broad word (many contexts) if all of the small word’s contexts are contained within the large word’s context set. But not vice versa.

4.2 4.2 Why Asymmetry Matters

Substitution itself is asymmetric. When “dog” can substitute for “animal” in some contexts, *all* contexts where “dog” appears are a subset of contexts where “animal” appears—but not vice versa. The containment score captures exactly this subset relationship.

In ECAN, HebbianLinks are symmetric. The parser suggests a concrete alternative: replace symmetric HebbianLinks with asymmetric *containment links*, where strength from A to B measures

$|\text{contexts}(A) \cap \text{contexts}(B)| / |\text{contexts}(A)|$. This would naturally capture taxonomic relationships (dog $\rightarrow$ animal has high containment; animal $\rightarrow$ dog has low containment) without requiring explicit taxonomic knowledge.

4.3 4.3 Connection to Cognitive Distance Research

Research on semantic distance in cognitive science has shown that conceptual distance is fundamentally asymmetric and organized by multiple dimensions. The parser operationalizes this: cognitive distance between two items is the *difficulty of substituting one for the other in context*, measured by containment scoring. This is not a pre-specified distance in a vector space; it is an empirically grounded, context-dependent, asymmetric measure.

5 5. Hierarchical Successive Generation: CKY as Attention Allocation

5.1 5.1 The Bottom-Up Process

The CKY loop processes spans shortest to longest. For each span, all binary splits are tested. At each split, the system retrieves cached substitution classes for left and right constituents, runs mutual expansion, forms the Cartesian product, validates against the corpus, scores valid combinations, and caches results for longer spans.

5.2 5.2 This IS Attention Allocation

This implements attention allocation more principally than ECAN in several respects:

1. **Bottom-up compositionality:** ECAN spreads importance along HebbianLinks without regard to compositional structure. The parser allocates attention in principled order: shorter spans first, because they are prerequisites for longer spans.
2. **Evidence-based importance:** In ECAN, initial importance is assigned by stimulus then spread. In the parser, importance is computed from corpus evidence—no external stimulus assignment needed.
3. **Self-limiting:** ECAN requires external rent mechanisms to prevent importance inflation. The parser’s energy function is naturally self-limiting: $+\log(\text{consensus})$ penalizes over-constrained classes, $-\log(n_{\text{full_length}})$ rewards structural productivity.
4. **Multi-scale attention:** Per-word-length truncation ensures substitutes of different lengths are preserved separately, preventing any single scale from dominating—maintaining attention at multiple levels of abstraction simultaneously.

6 6. Continual Learning via Corpus Fallback

6.1 6.1 The Two-Tier Architecture

Tier 1: UnitCatalog (pre-computed). Pre-indexed context patterns for high-frequency n-grams with GPU tensors for fast batch similarity. This is “compiled knowledge.”

Tier 2: CorpusIndex (runtime). On-demand access to raw corpus via n-gram index (up to 7-grams, no frequency threshold), with LRU caching and usage tracking for frequently accessed patterns.

The `get_unit_with_fallback` function bridges these tiers: try catalog first, fall back to corpus.

6.2 Mapping to AtomSpace Architecture

- **UnitCatalog $\sim$ Local AtomSpace**: Pre-indexed, GPU-accelerated, consolidated patterns for fast cognitive processing.
- **CorpusIndex $\sim$ Distributed AtomSpace / Raw Data Store**: Persistent storage queried when local AtomSpace lacks a needed pattern. LRU cache corresponds to working memory: frequently needed patterns promoted to fast local access.

The key insight for Hyperon: the parser demonstrates *how* to decide when to fall back to raw data. Whenever a Cartesian product combination is not in the catalog, the system checks the corpus. Novel combinations can still be evaluated if they appear in raw perceptual data. This is a concrete model for continual learning: handling genuinely novel inputs by accessing perceptual history, not just pre-compiled abstractions.

6.3 Dynamic GPU Index Extension

The `add_unit_to_gpu_index` method shows how newly discovered patterns are promoted to the fast tier—analogueous to adding new Atoms to a live AtomSpace. The knowledge graph grows during processing. New patterns discovered during reasoning are immediately available for subsequent reasoning steps within the same cognitive cycle.

7. Compositional Verification and Multi-Component Scoring

When forming Cartesian products of left $\times$ right substitutes, each combination is validated against the corpus. Only corpus-attested combinations survive. This is fundamentally different from Hyperon’s approach where PLN (Probabilistic Logic Networks) computes composite truth values from component truths via logical rules.

The parser’s approach is more grounded: the truth of “borrow your towel” as a substitute for “found my hat” is not deduced from component truths; it is verified by checking whether “borrow your towel” actually appears in the corpus with compatible contexts.

The multi-component scoring after verification includes context path counting (bridging paths between left and right substitutes), inner boundary diversity, outer context diversity, and outer context consensus—a rich, multi-dimensional evaluation that goes beyond simple truth-value composition.

8. Synthesis: A Proposal for Energy-Based Hypergraph Knowledge Representation

Drawing these threads together:

1. **Replace static truth values with dynamic energy functions.** Compute energy for *structural configurations*—proposed decompositions of complex Atoms. Minimum-energy decomposition defines the preferred interpretation.
2. **Replace symmetric HebbianLinks with asymmetric containment links.** Link strength from $A \rightarrow B$ measures what fraction of A ’s contexts are contained in B ’s. Naturally captures taxonomic relationships without explicit ontological engineering.
3. **Implement attention allocation as bottom-up compositional energy minimization.** Compute importance for compositional structures using an energy function rewarding substitutional diversity and empirical support.
4. **Maintain two-tier knowledge stores with dynamic promotion.** Pre-compiled AtomSpace for fast processing, with fallback to raw data for novel combinations. Dynamically add newly discovered patterns to the fast tier.
5. **Verify compositions empirically.** Check whether composed structures have grounding in observed data, not just logical consistency.
6. **Maintain multi-scale attention budgets.** Allocate attention separately at different hierarchical levels, preventing any single scale from dominating.
7. **Let distance and objects emerge from the process.** Do not pre-specify a distance metric or fixed conceptual categories. Let substitution classes (the “objects”) and the energy landscape (the “distances”) emerge from corpus-grounded energy minimization. This is the deepest insight: the structure of knowledge is not an input to the reasoning process but its product.

References

- SingularityNET. *MeTTa in a Nutshell: Exploring the Language of AGI*. <https://singularitynet.io/metta-in-a-nutshell-exploring-the-language-of-agi/>
- SingularityNET. *Hyperon Progress: From Prototypes to Scalable Intelligence*. <https://singularitynet.io/hyperon-progress-from-prototypes-to-scalable-intelligence/>
- SingularityNET. *OpenCog Hyperon Research*. <https://singularitynet.io/research/opencog-hyperon/>
- OpenCog Wiki. *AtomSpace*. <https://wiki.opencog.org/w/AtomSpace>
- OpenCog Wiki. *Attention Allocation*. https://wiki.opencog.org/w/Attention_allocation
- Goertzel, B. et al. *Economic Attention Networks*. AGI-09. https://agi-conf.org/2009/papers/paper_63.pdf
- OpenCog Wiki. *HebbianLink*. <https://wiki.opencog.org/w/HebbianLink>
- Deep Funding. *Framework for Evaluating Approaches to Attention Allocation*. <https://deepfunding.ai/rfp/framework-for-evaluating-approaches-to-attention-allocation/>
- Goertzel, B. et al. *OpenCog Hyperon: A Framework for AGI at the Human Level and Beyond*. arXiv:2310.18318. <https://arxiv.org/html/2310.18318>
- Kennington, C. et al. *What is Semantic Distance? A Review and Proposed Method*. <https://www.researchgate.net/publication/358937735>